

Solution Architecture

Date	15 July 2026
Team ID	*****
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

High-level architecture, mapped to Presentation, Application, and Data layers for the Smart Lender application.

Layer	Components
Presentation / Client Layer	Web application consisting of home.html (Home Page), predict.html (Loan Application Form), and submit.html (Prediction Result Page). The pages are rendered using Flask's render_template() and designed with HTML5 and CSS3.
API / Routing Layer	Flask routes handle user requests: GET / (Home Page), GET /predict (Loan Application Form), and POST /submit (Processes form data and displays prediction results).
Application / Core Logic Layer	Receives applicant information, performs preprocessing (encoding and scaling), loads the trained Random Forest model , predicts loan approval status, and formats the prediction result.
Data Layer	Uses loan_prediction.csv for model training and smart_lender_model.pkl (Pickle file) to store the trained Random Forest model. No runtime database is used in the current version.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Displays the homepage, loan application form, and prediction result page while collecting applicant information.	HTML5, CSS3, Jinja2
API Gateway (Flask routes)	Handles HTTP requests, accepts user input, invokes prediction logic, and returns the prediction result to the browser.	Flask, Python

Core Logic Service	Performs preprocessing (Label Encoding, Standard Scaling), loads the trained model, predicts loan approval, and prepares the output.	Python, scikit-learn, pandas, numpy
Model Store	Random Forest model trained using the loan prediction dataset to determine loan approval status.	Pickle (smart_lender_model.pkl)
Machine Learning Model	Random Forest model trained using the loan prediction dataset to determine loan approval status.	RandomForestClassifier (Scikit-learn)
Dataset	Contains applicant information used to train the machine learning model.	CSV (loan_prediction.csv)

High-Level Architecture Flow (For Report/PPT)

